

Ethical Hacking

Common Assignment

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1. Penetration Testing & Cybersecurity Assessment of a University Smart - Campus Infrastructure.

2. Introduction :-

A university Smart Campus Infrastructure contains many interconnected systems such as student web portals, Windows applications servers, Linux database servers, administrative systems, file servers, & wireless networks. These systems contain important information including student records, staff information, financial data, examination details, & research data.

Because these systems are interconnected, a vulnerability in one system may provide an attacker with an opportunity to access other systems. Therefore, penetration testing is required to identify weaknesses & recommend appropriate security controls.

Objective :-

- To identify Active hosts in the university network.
- To identify open TCP & UDP ports
- To enumerate Services running on Windows & Linux
- To identify potential vulnerabilities in Network Services
- To Assess the Security of the Student-facing web application.

Selected Cybersecurity Dataset :-

1. Denial of Service attacks

• Brute force attacks

• Port Scanning

• Web attacks

• Network reconnaissance

4. Attack Pattern Analysis :-

4.1 :- Port Scanning :-

Port Scanning is used to identify open Network ports & Services.

for example, an attacker may scan a university server to determine whether SSH, HTTP, SMB or other services are available

Impact:-

- Reveals the attack surface
- Helps attackers identify possible entry points.
- May expose unnecessary services.

Security Controls:-

- Strong passwords
- Multi-factor authentication.
- Account lockout / rate limiting

4.3 Denial of Service Attacks

→ DoS attack attempts to make a service unavailable by overwhelming its resource.

Impact on university:

• Student portal become unviable

- Online examinations may be interrupted
- Administrative services may become unavailable.

Security Controls:

- firewall filtering
- Rate limiting
- DDoS protection
- Network monitoring

Web Attacks

Web Application may be targeted through

Vulnerabilities such as

- Injection
- Broken authentication
- Broken Access Control
- Insecure session management

5. CIA Impact Analysis :-

The Vulnerabilities are evaluated using CIA triad

Confidentiality:-

Confidentiality protects information from unauthorized access.

University information requiring protection includes:

- Student personal information
- Examination records
- Financial information
- Staff information
- Research data

A Compromised web Application or database could result in Authorized disclosure.

Integrity:-

Integrity ensures that information cannot be modified unauthorized disclosure

Availability:-

Availability ensures that Authorized users can access services when required.

A successful DoS attack could make the student portal database.

UDP Port Scanning:-

UDP services can also be tested in the authorized environment

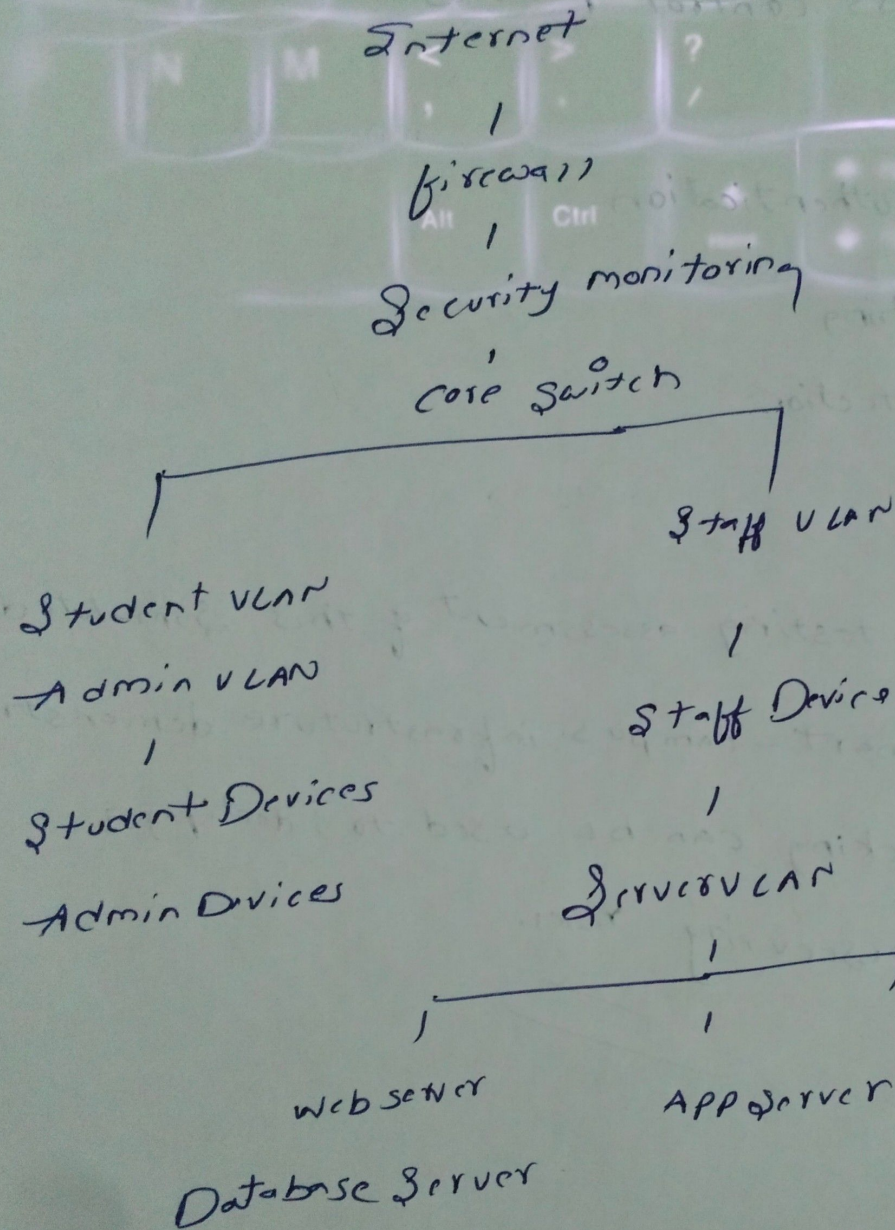
Ex:

- 30 192.168.56.40

Potential services include DNS, NTP, SNMP or other UDP-based services.

Hardened Smart Campus

Architecture



The Wireless network should also be separated into appropriate VLANs

Recommendation

- VLAN - based Network Segmentation
- Firewall & Access control policies
- HTTPS/TLS
- Multifactor authentication
- Regular Patching
- END Point protection.

Conclusion :

The penetration - testing assessment of this simulated university smart-campus infrastructure demonstrated how ethical hacking can be used to identify & reduce cybersecurity risks.